Coordinates & Transformations

Year 4 · **Homework Worksheet**

Worked Example

Worked example: A point is at (6, 2). It is translated 2 left and 5 up. What are its new coordinates?

Answer: 2 left: 6 − 2 = 4. 5 up: 2 + 5 = 7. New point: (4, 7).

Easy / Standard (15 questions)

1. A point A is at (7, 3). What is its x- coordinate?
2. A point A is at (7, 3). What is its y- coordinate?
3. A point starts at (2, 4) and is translated 3 right. What are its new coordinates?
4. A point starts at (1, 1) and is translated 5 up. What are its new coordinates?
5. A point is at (4, 1). It is translated 3 right. What are its new coordinates? A) (7, 1) B) (4, 4) C) (1, 1) D) (7, 4)
6. Point B is at (0, 8). Which axis does it lie on?
7. Point C is at (5, 0). Which axis does it lie on?
8. A point starts at (3, 3) and is translated 2 right and 2 up. What are its new coordinates?
9. A point starts at (6, 4) and is translated 2 left. What are its new coordinates?
10. Select ALL that apply. Look at the points P(5, 1), Q(5, 9), R(2, 1) and S(8, 9). Which have an x- coordinate of 5? A) P(5, 1) B) Q(5, 9) C) R(2, 1) D) S(8, 9)
11. A point starts at (1, 0) and is translated 5 right and 5 up. What are its new coordinates?
12. A point starts at (6, 6) and is translated 4 down. What are its new coordinates? A) (6, 2) B) (2, 6) C) (10, 6) D) (6, 10)
13. A point D is at (4, 2) and a point E is at (9, 2). Are D and E on the same vertical line or the same horizontal line?
14. A point starts at (2, 2) and is translated 5 right and 1 up. What are its new coordinates?
15. Which of these points lies exactly on the y-axis? A) (0, 5) B) (5, 0) C) (3, 3) D) (4, 4)

Standard (15 questions)

1. A point starts at (4, 2) and is translated 3 right and 3 up. What are its new coordinates?
2. A point is at (6, 5). It is reflected in the x-axis. What are its new coordinates?
3. A point starts at (7, 8) and is translated 5 down. What are its new coordinates?
4. A point at (3, 7) is reflected in the x-axis. What are its new coordinates? A) (3, -7) B) (-3, 7) C) (7, 3) D) (-3, -7)
5. A point is at (4, 6). It is reflected in the y-axis. What are its new coordinates?
6. A rectangle has vertices at (0, 0), (7, 0), (7, 2) and (0, 2). What is the length of its longer side?
7. A point starts at (5, 3) and is translated 2 right and 6 up. What are its new coordinates?
8. A point starts at (3, 9) and is translated 5 left. What are its new coordinates?
9. A point at (6, 3) is reflected in the y-axis. What are its new coordinates? A) (-6, 3) B) (6, -3) C) (3, 6) D) (-6, -3)

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10. A point starts at (0, 6) and is translated 4 right and 1 down. What are its new coordinates?

11. Select ALL that apply. A square has vertices at (1, 1), (1, 5), (5, 5) and (5, 1). Which statements are true? A) The square has a side length of 4 units B) The square has a side length of 5 units C) The point (5, 5) is a vertex of the square D) The point (1, 5) is a vertex of the square

12. A point starts at (3, 4) and is translated 3 right and 3 up. What are its new coordinates?

13. A point is at (8, 2). It is reflected in the x-axis. What are its new coordinates?

14. A point starts at (4, 4) and is translated 3 left and 4 up. What are its new coordinates? A) (1, 8) B) (7, 8) C) (1, 0) D) (7, 0)

15. A point starts at (6, 0) and is translated 5 up. What are its new coordinates?

Standard / Advanced (15 questions)

1. A point is at (5, 8). It is reflected in the x-axis, then translated 3 up. What are its final coordinates?

2. A rectangle has vertices at (0, 0), (7, 0), (7, 3) and (0, 3). What is the perimeter of the rectangle? A) 10 B) 14 C) 20 D) 21

3. A point at (6, 4) is reflected in the y-axis. What is the sum of its new x-coordinate and y-coordinate?

4. A point is at (3, 3). It is translated 5 right and 5 up, reaching point Q. Is Q nearer to the x-axis, nearer to the y-axis, or the same distance from both?

5. A square has vertices at (1, 1), (1, 6), (6, 6) and (6, 1). What is its perimeter?

6. A point is at (9, 4). It is translated 3 left and 2 down, reaching point Q. What are its coordinates?

7. A rectangle has vertices at (0, 0), (8, 0), (8, 3) and (0, 3). What are the coordinates of the vertex diagonally opposite (0, 0)? A) (8, 0) B) (0, 3) C) (8, 3) D) (3, 8)

8. Take the point (5, 8) and reflect it in the x-axis. Where does it land?

9. A square has vertices at (2, 2), (2, 8), (8, 8) and (8, 2). What is its perimeter?

10. Reflecting the point (9, 4) in the y-axis gives which result? A) (-9, 4) B) (9, -4) C) (-9, -4) D) (4, 9)

11. A point starts at (2, 6) and is translated 6 right and 6 up. What are its new coordinates?

12. Select ALL that apply. A rectangle has vertices at (2, 1), (7, 1), (7, 5) and (2, 5). Which statements are true? A) Its width (horizontal side) is 5 units B) Its height (vertical side) is 4 units C) The vertex (7, 5) is diagonally opposite (2, 1) D) All four vertices have an x-coordinate greater than 10

13. A shape has vertices A(0, 2), B(0, 6), C(3, 6) and D(3, 2). It is translated 3 right and 4 up. What are the new coordinates of A and B?

14. A point starts at (0, 0) and is translated 3 right and 4 up, reaching point Q. What are the coordinates of Q?

15. Suppose the point (6, 4) is moved 6 units to the left and 4 units downward. Of the four options below, which one correctly describes where it finishes up? A) Exactly on the origin, (0, 0) B) Somewhere on the x-axis, but not at the origin C) Somewhere on the y-axis, but not at the origin D) Still within the same quadrant as before

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